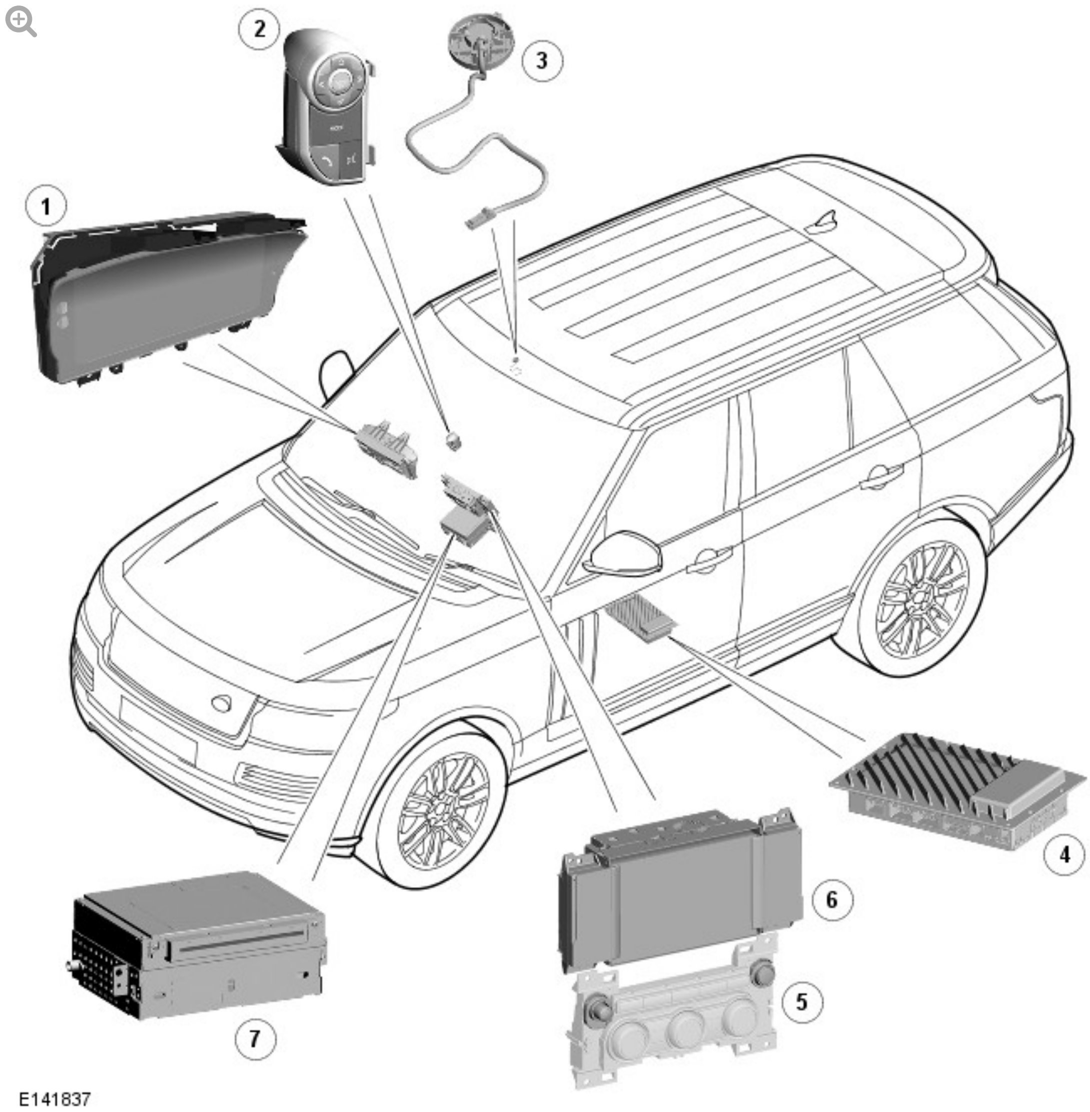
INFORMATION AND ENTERTAINMENT SYSTEM

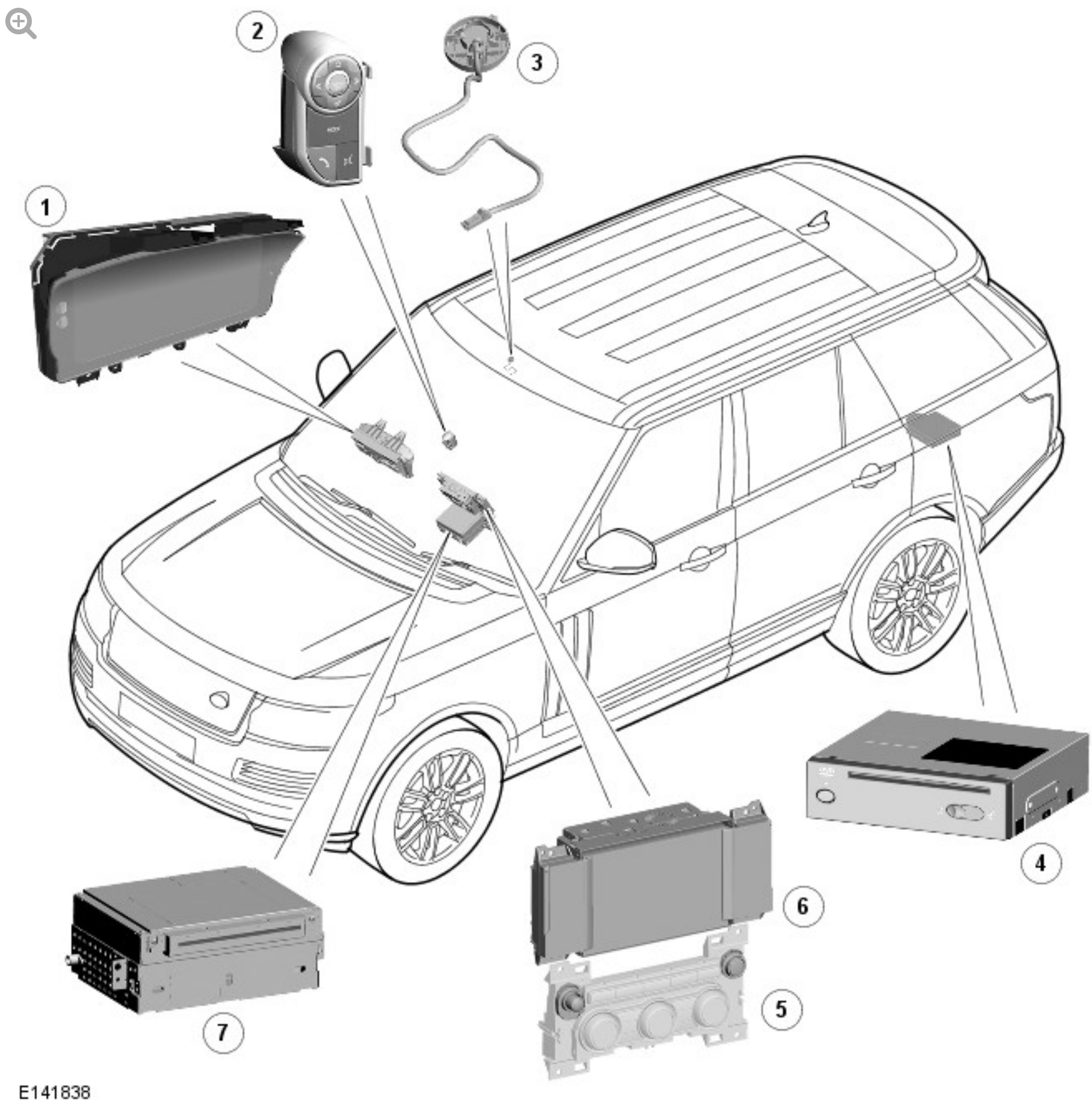
DESCRIPTION AND OPERATION

COMPONENT LOCATION - ROW



ITEM	DESCRIPTION
1	Instrument cluster
2	Left steering wheel switchpack
3	Microphone
4	Integrated Control Panel (ICP)
5	Touch Screen (TS)
6	Integrated Audio Module (IAM)

COMPONENT LOCATION – JAPAN/ASIA



ITEM	DESCRIPTION
1	Instrument cluster
2	Left steering wheel switchpack
3	Microphone
4	Navigation control module
5	Integrated Control Panel (ICP)
6	Touch Screen (TS)
7	Integrated Audio Module (IAM)

OVERVIEW

NOTE:

Only basic voice controls are available for Japan specification vehicles. Voice control is not available in other Asian markets.

The Rest Of World (ROW) Advanced Range Rover Voice system provides the driver with the option of voice control for a range of supported functions. In addition to the navigation system, phone system and the notepad functions, the system also supports radio, satellite radio, digital radio, single CD (compact disc), virtual CD changer, USB (universal serial bus) and auxiliary connection functions (where fitted).

The ROW Advanced Range Rover Voice system adopts a concept known as 'Say What You See' (not applicable to Japan specification vehicles). Each of the Advanced Range Rover Voice functions are supported by 'Help' commands; saying 'Help' at each point in the conversation will give a context sensitive explanation of what the user can do at that point.

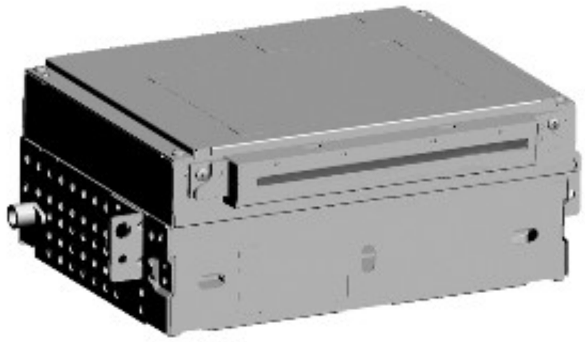
The system allows the vehicle user to concentrate fully on driving the vehicle, without any need to divert their eyes from the road ahead in order to check information read outs on the vehicle instrument panel or Touch

Screen (TS). The voice control system also feeds back audible information to the vehicle user.

Voice control is mainly a software based system. The software for controlling the voice system is resident in the Integrated Audio Module (IAM), TS and Instrument cluster. The IAM and TS each contain more than one voice control software component and communication between all three modules is via the Media Oriented System Transport (MOST) ring. A voice control microphone is located in the front overhead console and is hardwired to the IAM.

DESCRIPTION

INTEGRATED AUDIO MODULE (IAM)



E141841

NOTE:

The Japanese specification voice system does not store map and voice data on the IAM. All other functions of the IAM are applicable to Japan specification vehicles. Refer to 'Japanese Voice System' section below for details of the Japanese voice system.

The IAM is located in central position in the instrument panel, behind the Integrated Control Panel (ICP).

The IAM is connected on the MOST ring to the other infotainment system components. The hands free microphone is also hardwired to the IAM. The microphone is used for voice control, hands free telephony and Dynamic Volume Control (DVC). The IAM contains the recognizer, dialogue and Text-to-Speech (TTS) parts of the voice engine used to determine what the user wants to control by voice.

A 40 GB hard drive is used primarily for storing the information for satellite navigation. A 10GB partition is provided for storing music files, the remaining 30GB is used for map and voice control data storage e.g. Voice feedbacks.

The integral hard drive for the navigation system removes the requirement of a separate navigation computer usually found in the rear luggage compartment. The IAM stores the navigation map and voice control data locally within the 30GB hard drive partition. Map upgrades and software now have to be loaded directly into the IAM from a CD (compact disc).

The IAM communicates on the MOST ring with the rest of the Infotainment system. If the IAM is replaced it must be configured as a new module using an approved Range Rover diagnostic system. Calibration of the IAM using an approved Land Rover diagnostic system enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

TOUCH SCREEN (TS)

E141842

The Touch Screen (TS) is mounted centrally in the instrument panel. It provides the user interface to the rest of the Infotainment system. As well as being connected to other vehicle systems via the medium speed CAN (controller area network) comfort bus and the rest of the infotainment system via the MOST ring, the left steering wheel switchpack for infotainment control are also connected to the TS via a hardwired resistive ladder.

The controlling software for the voice system is contained in the TS. This is responsible for starting and ending the voice engine in the IAM as well initiating any requested actions. It also provides information to a 'Say What You See' visual prompter in the TS which determines what to show in the instrument cluster during a dialogue with the voice system.

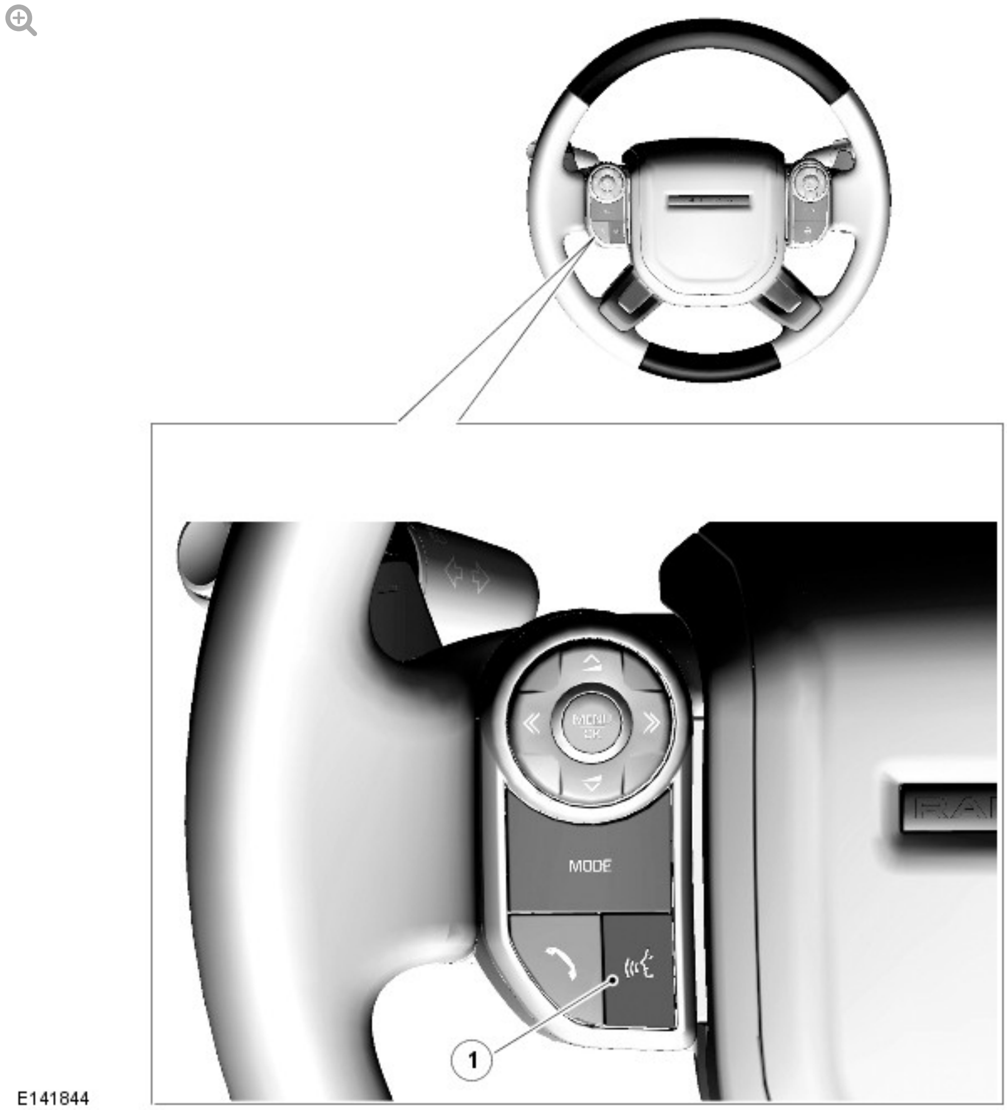
The driver can control voice functions by briefly pressing the voice button on the left steering wheel switchpack. The TS will then initiate a voice session in the IAM, allowing the user to say voice commands to activate some of the infotainment system features.

MICROPHONE

E141843

The directional type microphone is fitted to the driver side of the overhead console so that it is directed towards the driver. It is connected to the IAM for hands free telephony, voice control and dynamic volume control (DVC) systems. The IAM has an integrated noise suppression and echo cancellation system for hands-free telephone use.

LEFT STEERING WHEEL SWITCHPACK



ITEM	DESCRIPTION
1	Voice switch

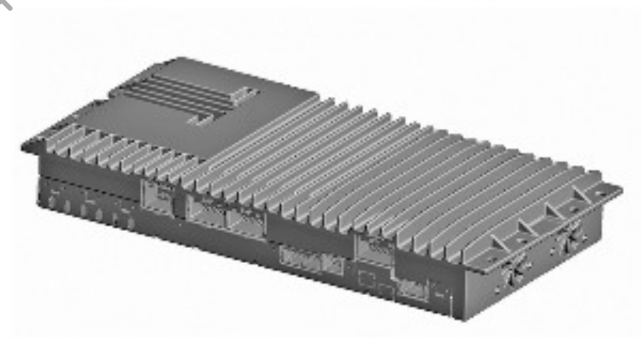
The push-to-talk switch for the voice system is located on the left steering wheel switchpack. The driver is able to use this switch for:

- Starting a voice session
- Cancelling/Ending a voice session
- "Barging in (Interrupting) during a voice session (stops playing the current feedback and moves directly to the next listening tone, allowing the user to immediately enter the next command rather than having to wait).

AUDIO AMPLIFIER MODULE

NOTE:

Meridian Signature Reference System audio amplifier module shown



E141845

The audio amplifier module is located below the front left seat. It is connected to the audio system via the MOST bus. Speaker connections from the amplifier are hardwired.

JAPANESE VOICE SYSTEM

The Japanese voice system is part of the Japanese navigation system and uses the standard system components, with the exception that the map and voice data is not stored on the IAM hard disc drive. Additional components are:

- a navigation control module
- a navigation video interface module.

The two additional modules are used to read the map and voice data and output audio and video signals to the TS, IAM and audio amplifier module.

NAVIGATION CONTROL MODULE

A separate navigation control module is located on the floor of the luggage compartment.

E141846

The navigation control module is a DVD drive which reads voice and map data direct from a DVD. The navigation control module is connected on the MOST ring and communicates with the TS to initiate a voice session.

The navigation control module outputs the video signals in a Gigabyte Video InterFace (GVIF) format to a video interface module which converts the GVIF input to a Low-Voltage Differential Signalling (LVDS) video signal output which is then passed to the TS. Audio output is on the MOST ring to the audio amplifier module.

VICS FM transmission signals are received by the navigation control module via an FM antenna and a VICS antenna amplifier. Infra-red VICS transmissions are also received by the VICS beacon antenna, located on the top of the instrument panel, and are passed to the navigation control module.

OPERATION

NOTE:

> Only basic voice controls are available for Japan specification vehicles.

The ROW advanced voice system provides the driver with the option of voice control for a range of supported functions. The following systems can be controlled by Voice:

- Bluetooth® Telephone system
- Navigation system
- Vehicle notepad
- Radio, satellite radio and digital radio (if fitted)
- CD player
- Virtual CD player
- USB and auxiliary connections.

The 'notepad' facility allows voice notes to be recorded. Voicetags for phone dialling, radio stations and navigation locations allow the system to be personalized and there are help and tutorial functions to provide advice on using the system.

The Advanced Voice system uses a 'visual prompter' system known as 'Say What You See' (SWYS). Each of the Advanced Voice functions are supported by 'Help' commands; saying "Help" at any point in the conversation will give a context sensitive explanation of what the user can do at that point. The SWYS visual prompter shown in the instrument cluster always guides the user through the flow, showing not only examples of what they can say next, but also confirmation of where they are in the conversation flow.

Steering Wheel Controls and Instrument Cluster View

ITEM	DESCRIPTION
1	Voice button
2	Voice symbol
3	SWYS visual prompter
4	When displayed, say Cancel to cancel the current voice session
5	When displayed, say Help to get assistance during a voice session

NOTE:

Telephony voice commands are only accessible when a Bluetooth® device is docked to the vehicle.

The system is operated by the voice button on the left steering wheel switchpack. Voice commands are detected by the dedicated microphone in the overhead console. When giving a voice command audible feedback will be heard through the vehicle's audio speakers.

Efficient operation of Advanced Voice Control is reliant on the user understanding some of the following basic operating conditions;

- Face forwards, sitting in a normal driving position

- After pressing the voice button, always wait for the end of the audible tone before speaking
- Speak naturally, as if you were talking to a passenger or on the phone without pausing between words
- When the system asks for more information, always wait for the end of the tone before responding
- Always say numbers clearly but at a natural pace
- Excessive noise, for example while driving with windows open, may cause voice command mis-recognition. If it is too noisy to use the phone, it is likely that voice commands will not be recognized.

Most accents are understood without difficulty, but if the system does not recognize the command it will respond "SORRY" and allow two more attempts to say the command.

Voice feedback is given in the same language as the command recognition. It is possible to change the language of the speech control system by changing the system language.

COMMAND LIST

A full list of all shortcut voice commands can be found on the Touch Screen (TS) by pressing the 'command list' softkey in the 'Voice Settings' menu. Select a device name from the list to see all the shortcut commands available to interact with that device. Pressing the '**i**' softkey next to any command shows alternative ways of saying the same command.

STARTING A VOICE SESSION

To start a voice session the driver operates the voice switch on the left steering wheel switchpack briefly. An audible tone can be heard, followed by the presentation of the 'Say What You See' (SWYS) visual prompter in the instrument cluster indicating some commonly used available commands. A voice symbol alongside each item in the list indicates that the system is listening. Always wait for the listening tone to finish playing before using the command. To end a session the same button is pressed and held until a double beep is heard.

When the voice switch is operated on the left steering wheel switchpack, a voltage is received at the TS via the clockspring assembly. This voltage is sent on a single wire from the switchpack, through a resistive ladder. The whole process is then initiated via the MOST network; for example the TS starts the voice session and carries out the resulting action requested by the user, but the IAM maintains the dialogue with the user. The SWYS visual prompter is calculated in the TS, but presented in the instrument cluster when requested to do so by the IAM. The accompanying voice feedback is sent to the audio amplifier module for broadcast over the speakers from the IAM. If a recognized user instruction is received via the microphone, this is then processed and sent to the TS to perform the required action.

NOTE:

> Should an instrument cluster priority message be required, this will prevent or cancel the current voice session.

VOICE TUTORIAL

There is a voice tutorial that can be accessed either by giving the 'voice tutorial' command or by pressing the 'voice tutorial' softkey located in the 'Operating Guide' option in the 'Voice Settings' menu on the TS. The tutorial gives information on how to use the voice system, useful tips and how an experienced user can speed up their interaction with the voice system by turning voice feedback off, using barge-in (interrupt) and saying shortcut commands.

VOICETAGS

Voicetags allow the user to store voice entries as shortcuts to control various functions; for example routing to navigation locations, dialling numbers and tuning to radio stations. The voicetags sub-menu, in the voice settings menu in the TS, accesses controls for navigation, phone, radio and depending on specification digital radio or satellite radio.

VOICE TRAINING

Voice training is used to adjust the system to recognize the user's voice more accurately. The voice system allows two different users to create separate profiles, providing training for User 1 and User 2. When training is activated for each user, a 'pop-up' is displayed to confirm that training is in process for that user. The 'pop-up' informs the user that voice training must be fully completed in order to activate the new voice profile, and offers the option of 'OK' to initiate the session and store data in that User profile, or 'Cancel' to return to the previous menu. Once activated, another 'pop-up' indicates that training is in progress.

Voice training phrases will be shown on the 'SWYS visual prompter' in the Instrument Cluster and the user will be requested to say each phrase after the listening tone.

NOTE:

> Voice training can only be conducted stationary with the engine running and with the climate control NOT in defrost due to background noise.

Voicetags and training are stored in a non-volatile memory within the IAM. Disconnection of the battery would not cause any customer data loss.

NOTES:

- To enable new voicetags and training to be written to memory, a period of ten minutes after the last key off cycle must take place. Should the battery be disconnected before this time then data may be lost.
- If the IAM was to be replaced then all voicetags and training would be lost.
- If either the IAM, Instrument Cluster or the TS are replaced, it is recommended that the vehicle language settings and voice language settings (if vehicle language is not supported by voice control) are reset to the same setting.

NAVIGATION DESTINATION ENTRY BY VOICE

Destination entry uses phonetic transcriptions of the navigation data (stored as part of the map data) to offer the user the ability to enter an address or postcode into the Navigation system by voice. The user simply follows the visual and audible instructions given by the voice system and enters their desired address in a step-by-step manner (e.g. city, then street, then house number).

At each address entry stage, the user's voice command is matched against the phonetic map data and a list of likely recognition candidates is presented in a "picklist" for the user to select from. If the chosen address has more than one location associated with it, the voice system will work with the user to determine the exact address they wish to navigate to.

DIALLING FROM THE G2P PHONEBOOK

Provided the phonebook has been downloaded via Bluetooth®, the voice system is able to perform a grapheme-to-phoneme (G2P) transcription of each of the names stored in the phonebook. This is then used by the voice system to allow the user to dial a contact by saying the name stored in the phonebook, there is no need to store a voicetag first.

The user's voice command is matched against the phonebook entries and a list of likely recognition candidates is presented in a "picklist" for the user to select from. If the chosen contact has more than one number associated with it, the voice system will work with the user to determine the exact number they wish to dial.

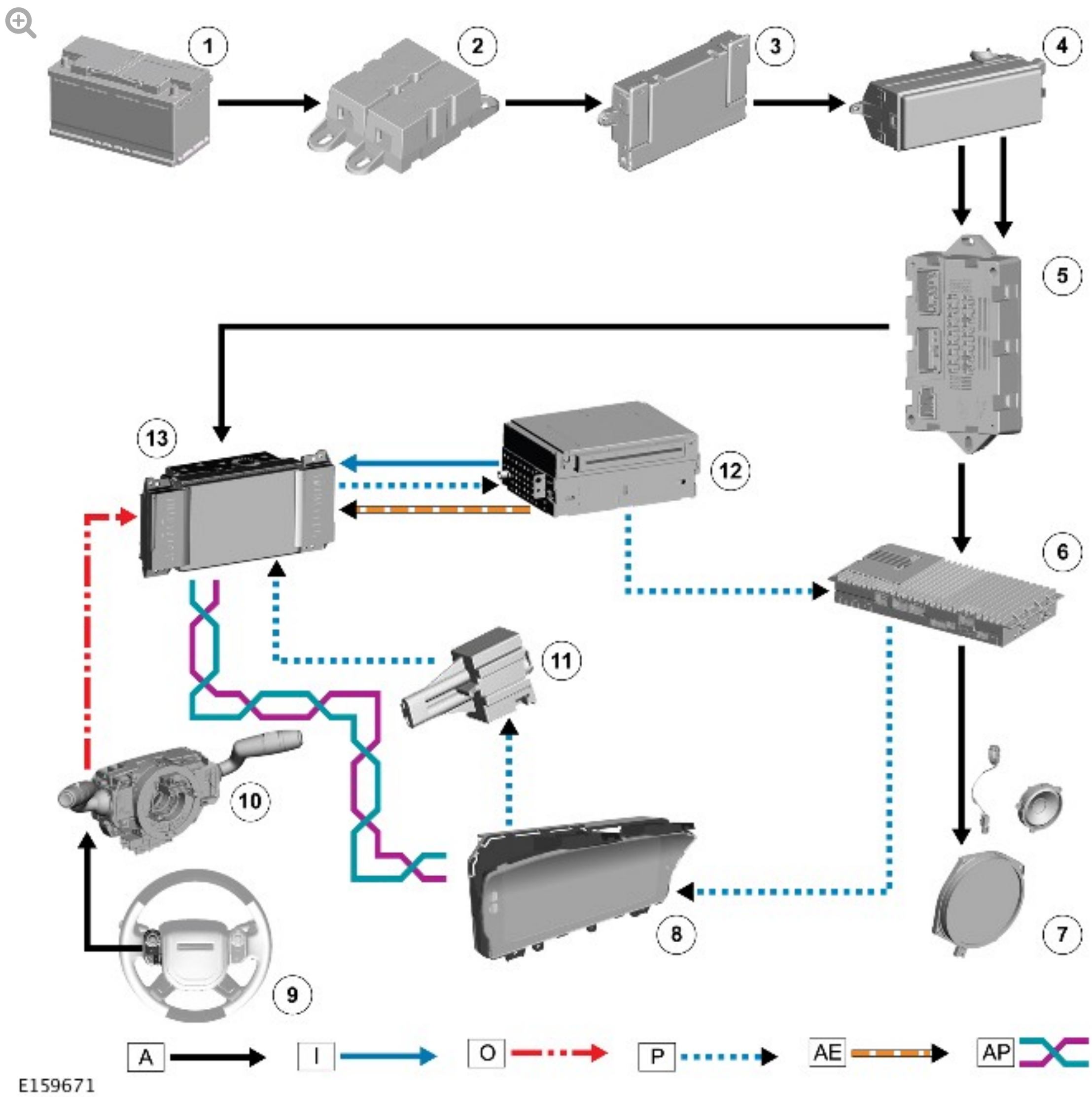
NOTES:

- The contact name must be spoken as displayed in the phonebook in the TS. For example, if "Smith Steven" is shown in the Phonebook in the TS, it is unlikely that the voice system will recognise "Steven Smith" (order reversed), "Steve Smith" or just "Steven" (partial name) correctly.
- The system will attempt to recognise names based on the currently selected voice language. Foreign contacts may not be recognised correctly and depending on the characters used, may not be recognised at all.
- For regularly used contacts with more than one number or a foreign name, the user can store a voicetag as a shortcut.

VOICE CALLER ANNOUNCE

When enabled in the Phone settings, the Voice system will announce the name of the incoming caller identity through the system speakers.

CONTROL DIAGRAM – ROW

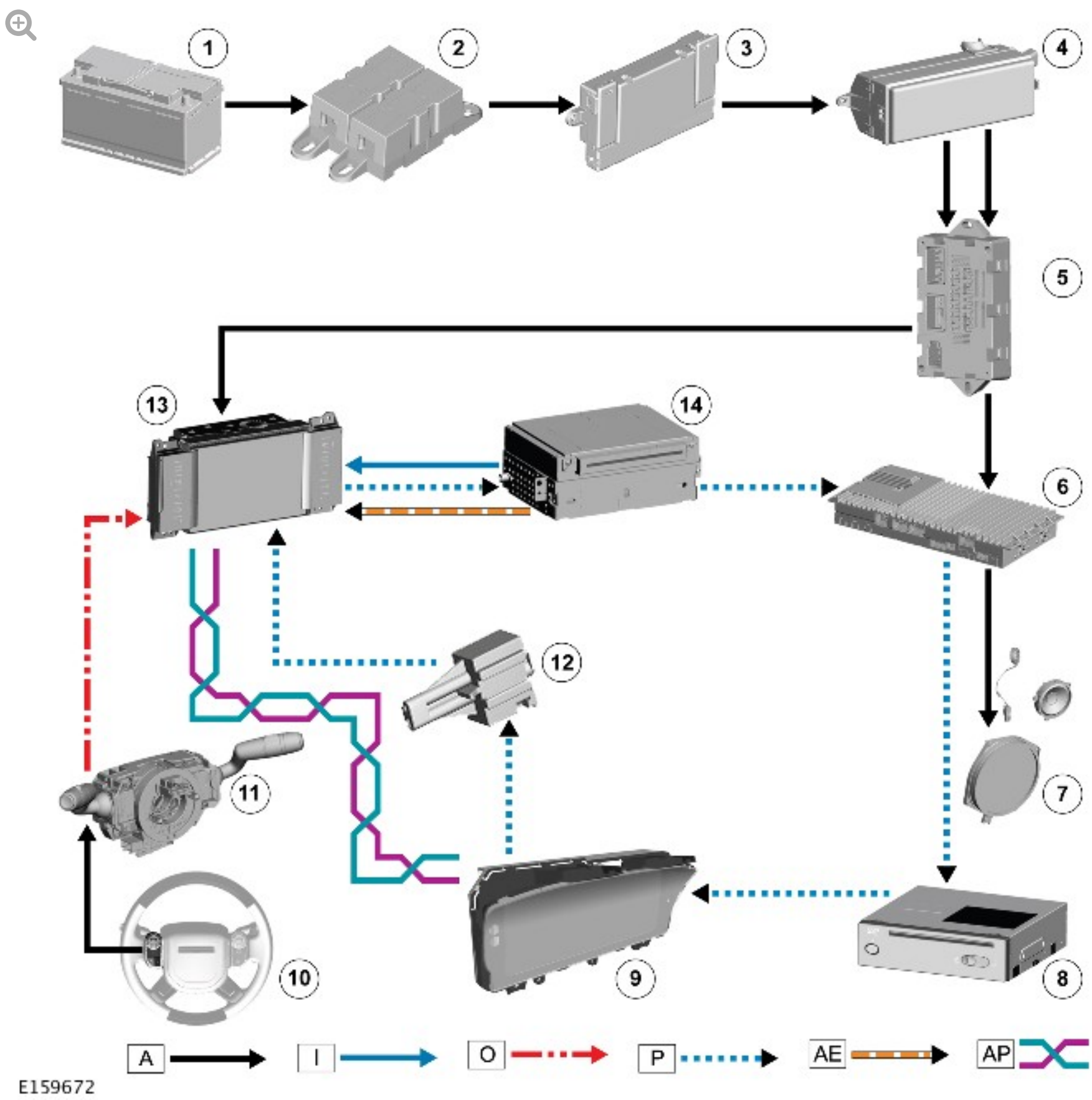


A = HARDWIRED; I = CVBS (COMPOSITE VIDEO BASEBAND SIGNAL); O = LIN (LOCAL INTERCONNECT NETWORK) BUS; P = MOST (MEDIA ORIENTED SYSTEM TRANSFER); AE = LVDS (LOW-VOLTAGE DIFFERENTIAL SIGNALLING); AP = MS (MEDIUM SPEED) CAN COMFORT BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Quiescent Current Control Module (QCCM)
6	Audio Amplifier Module (AAM)

7	Vehicle speakers
8	Instrument Cluster (IC)
9	Left steering wheel switchpack
10	Clockspring
11	MOST diagnostic connector
12	Integrated Audio Module (IAM)
13	Touch Screen (TS)

CONTROL DIAGRAM - JAPAN



A = HARDWIRED; I = CVBS (COMPOSITE VIDEO BASEBAND SIGNAL); O

= LIN BUS; P = MOST (MEDIA ORIENTED SYSTEM TRANSFER); AE = LVDS (LOW-VOLTAGE DIFFERENTIAL SIGNALLING); AP = MS (MEDIUM SPEED) CAN COMFORT BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Quiescent Current Control Module (QCCM)
6	Audio Amplifier Module (AAM)
7	Vehicle speakers
8	Navigation Control Module (NCM)
9	Instrument Cluster (IC)
10	Left steering wheel switchpack
11	Clockspring
12	MOST diagnostic connector
13	Touch Screen (TS)
14	Integrated Audio Module (IAM)